

Master Structure v2.1 — Pass 1, comprehensive

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How to read the master structure

This is the comprehensive version: every unit of the program, at every depth, with what it teaches, which artifact field it feeds, how it is delivered, and what it needs; the condensed versions are not separate documents but slices of these tables, selected by the Depth column.

Depth and slices. O = Orientation, P = Practice, M = Mastery. The Introduction slice is every O unit (about 45 h). The Practitioner slice is every O and P unit with one area (about 200 h). The Comprehensive version is everything (about 480 h with one studio and two tool electives). A single module at any depth is its own product. Change a unit here and every slice changes with it.

Delivery codes (a unit usually has two or three): L = lecture material (text, short video; §11 says what qualifies) · C = case, an object or project from the world set told as a story · I = interactive lab (single concept) · S = simulation (compound, several concepts coupled; §9) · P = the AI partner in one of its voices · K = physical making with things at hand · D = a document the learner produces · T = a tool unit is used here.

Media codes (§10 has the sourcing): img = images from open collections · drw = drawings, plans, sections · 3d = models · vid = short video · aud = audio · dat = data · ani = animated diagram · pack = a site, object, body or brand pack for a studio brief.

Artifact field. The Concept DNA section each unit writes to, by the codes in §2.

Research status. Every module is Open: research is uncapped and logged (§12). Status moves to Sprinted (one research sprint done), Drafted (units written), Tested (labs run with people), Live.

Versioning. This structure is v2.1. A module's research may add, split, merge, move or re-depth units; each such change increments the structure (v2.2, v2.3...) and is recorded in §12's change log. Nothing here is final by design.

Regional slot. Every module's last unit is a regional slot: the same concepts on the learner's own place, and the hook for the Gulf, MENA and SWANA vertical later.

Counts. 4 levels, 21 modules, 10 tools modules, 10 compound engines, about 430 units. Numbering is module.unit (F1.3, S2.14) so units can be referenced from anywhere.

The artifact schema

Every unit writes to one of these fields; the app holds the artifact as this structure, and the field codes appear in every unit table below.

Code	Field	Filled mainly by
SRC	Sources, attributions, consultation record	F1, M1, R2
PRM	Premise: proposition, stance, contradiction, scenario, non-goals	M1
PER	Perimeter: what the world enables, protects, resists, rejects, transforms	F5, M1
RUL	Rules: constants, variables, thresholds	M2
ANT	Anti-DNA	M2
GRM.f / .m / .k / .c / .s / .a / .sn / .b / .i / .t / .n / .v	Grammars: formal, material, construction (k), chromatic, spatial, atmospheric, sensory, behavioural, interaction, temporal, narrative, verbal	Candidates from F2–F4; rules in M2; refined in areas
ALN	Alignment with practice, collection and masterbrand DNA	M2, R1
OBL	Category obligations and permitted variation	M2
TST	Test set: belongs, edge, failure, seductive-but-off	M2, every module's closing test
BRF	Contributor brief and agent prompt architecture	M2, T3

Code	Field	Filled mainly by
VER	Version history and change log	M2 onward
COL	Collection map: roles, families, drops, carriers, distribution	M3
ANC	Anchor expression and family	M4, areas
ADJ	Adjacent translation	M4, areas
JDG	Judged set (briefer lane)	Every module's closing test, briefer lane
CMB	Commissioning brief and record (briefer lane)	Areas, R3
VEN	Venture: masterbrand, orbital architecture, community, campaign	R1
RGT	Rights and money: IP protocol, funding scenario, cash model	R2
HND	Handover package	R3, areas
LCY	Lifecycle: release, maintenance, repair, end of life	R4
PLW	Planetary weight ledger, accumulating from F5 to R4	F5, S-areas, T9, R4
DOS	Dossier: evidence, alternatives, abandoned directions, future practice	R5
SITE	The site, object, body or brand pack a studio brief runs on	Areas at Mastery

Level 0 Orientation and Level 1 Foundations

Foundations is 71 units across five modules; at Practice depth it is a complete global design-literacy course on its own, and its Mastery units are research and specification pieces rather than studios.

O1 How to see (4 units, all O)

Unit	Concept	Depth	Field	Delivery	Interactive · media
O1.1	What design does in the world: every made thing tells you what to do, what to feel, who you are	O	—	L, C	img
O1.2	Read a Room: annotate a space or object for its instructions	O	—	I	Read-a-Room annotator · learner's photo
O1.3	The artifact you will build, and the two lanes	O	—	L	ani
O1.4	Meeting the partner: its three voices, what it never does	O	—	P	—

F1 Histories of making (35 units, structure v3.6)

Unit	Concept	Depth	Field	Opening · forms	Voice · media	Was (v2.1)
F1.1	One object, three readings: material, labour, meaning; the Lineage view opens here	O	SRC	Object first · 3	Historian · img	new
F1.2	The reference-to-appropriation spectrum, read as a plane	O	SRC	Prediction first · 15, 3	Historian · img	F1.2
F1.3	How histories get written: canons, periods, the timeline as an argument; design history and design studies; the state-authored heritage list as a bias of the record	P	SRC	Argument first · 13, 11	Mirror · ani	F1.3
F1.4	Interdisciplinary discourse: what film, performance, sound, gaming and digital culture teach making	P	SRC, GRM.n	Story first · 10, 7, 8	Mirror · vid, aud	F1.3a (v3.0 addition)

Unit	Concept	Depth	Field	Opening · forms	Voice · media	Was (v2.1)
F1.5	Recurrences: writing invented three or four times, with Egypt as the contested case; cities, metallurgy, weaving and printing many times	P	SRC	Data first · 1, 11 (simultaneity view)	Historian · img, dat	F1.1 (the nine pivots)
F1.6	Worlds of making: Africa, read through metallurgy; ends on a living practice	P	SRC	Object first · 3, 10	Historian · img	F1.4
F1.7	Worlds of making: the Americas, read through fibre; ends on a living practice	P	SRC	Object first · 3, 10	Historian · img	F1.5
F1.8	Worlds of making: East Asia, read through the workshop; ends on a living practice	P	SRC	Object first · 3, 10	Historian · img	F1.6
F1.9	Worlds of making: South and Southeast Asia, read through cotton; ends on a living practice	P	SRC	Object first · 3, 10	Historian · img	F1.7
F1.10	The Islamic world as a network: routes and the waqf; ends on a living practice	P	SRC	Data first · 12, 3	Historian · img, drw (Archnet)	F1.8
F1.11	Europe as one world among several, read through the guild; ends on a living practice	P	SRC	Object first · 3, 10	Historian · img	F1.9
F1.12	Worlds of making: Oceania and the Pacific, read through navigation; ends on a living practice; the sensitivity notice explained	P	SRC	Object first · 3, 10	Historian · img	F1.10
F1.13	Networks of objects, materials and skills: Silk Roads, Indian Ocean, trans-Saharan, Mediterranean, Atlantic, Pacific; people never drawn as goods	P	SRC	Data first · 12, 1	Historian · dat, img	F1.11

Unit	Concept	Depth	Field	Opening · forms	Voice · media	Was (v2.1)
F1.14	Copying, transfer and counterfeit: porcelain, chintz, lacquer, print	P	SRC, RGT	Object first · 5, 3	Historian · img	new
F1.15	Ornament as a global language; the exhibition as taxonomy	P	GRM.f, SRC	Prediction first · 3, 10	Historian · public-domain plates	F1.12
F1.16	Ritual and belief as drivers of making	P	GRM.n, SRC	Object first · 3, 8	Historian · img, aud	new
F1.17	Who commissions: patrons, guilds, merchants, industry, agencies, platforms (the exemplar spec)	P	SRC, CMB	Argument first · 2, 13	Client · ani	new
F1.18	Textiles and the making called women's work; women's making beyond textiles	P	SRC, GRM.m	Making first · 3, 7	Historian · img	new
F1.19	Industry, the machine and the exhibition	P	SRC	Story first · 1, 3	Fabricator · img	F1.13
F1.19a	War, crisis and making: rationing, military production, reconstruction, propaganda, displacement	P	SRC, PER	Story first · 1, 12	Fabricator · img	new (v3.4)
F1.20	Modernisms, plural	P	SRC	Object first · 10, 13	Historian · img	F1.14
F1.21	Consumer society, the digital and the platform	P	SRC	Story first · 1, 7	Client · img, vid	F1.15, split
F1.22	The planetary present: making as extraction	P	SRC, PLW	Data first · 12, 4	Planet · dat	F1.15, split
F1.22a	Use and consumption: how things were lived with, worn out, repaired, resold, inherited	P	SRC, LCY	Object first · 3, 5	Mirror · img	new (v3.4)
F1.23	Obscured labour: the credit map	P	SRC	Object first · 3	Historian · img	F1.16
F1.24	The museum as author: collecting, provenance, restitution	P	SRC	Argument first · 13, 3	Historian · img	new

Unit	Concept	Depth	Field	Opening · forms	Voice · media	Was (v2.1)
F1.25	Reading an object as history: slow looking and the autopsy in five layers (material, technique, labour, trade, meaning); Practice Reading as its sibling	P	SRC	Lab first · Object Autopsy, full	Mirror · img	F1.17
F1.26	Attribution and consultation as protocol, on the CARE principles	P	SRC	Problem first · 2, 15	Historian · template	F1.18
F1.27	Counter-history research piece on one lineage	M	SRC, DOS	Artifact first · Atlas authoring	Historian · learner's material	F1.19
F1.28	Primary sources and archives: how to read them	M	SRC	Problem first · Source Grader	Historian · img	F1.20
F1.28a	Reconstruction: making to know; one technique rebuilt from a primary text, with what the text left unsaid	M	SRC	Making first · operational-sequence strip	Fabricator · primary text, img	new (v3.6)
F1.29	Vernacular intelligence: oral history, folktale, local taxonomy, with consent	M	SRC	Making first · Oral History Kit	Historian · aud	F1.20a (v3.0 addition)
F1.30	Re-timing the Atlas: other calendars and periodisations	M	SRC	Data first · Calendars	Historian · dat	new
F1.30a	Counterfactual history: a history of making that did not happen	M	SRC, PRM	Problem first · 1, 13	Stranger · —	new (v3.4)
F1.31	Regional slot: the learner's own place, authored locally	M	SRC	Artifact first · Lineage	Historian · learner's material	F1.21

F1 is the first module in the v3 row format. The Delivery column gives way to the opening move and the presentation forms (numbered as in Final Structure v3.2), the partner voice is named, and Was maps every v2.1 number, so the v3.0 additions F1.3a and F1.20a are now F1.4 and F1.29. Two units at Orientation, twenty-six at Practice, seven at Mastery; about 4 /

22 / 22 hours. Apply by lane, the closing test, the descriptor and the evidence for every change live in the F1 Module File; other modules move to this format when they pass gate G1.

F2 Form (15 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
F2.1	Archetype DNA: what makes a chair a chair	O	GRM.f	I	Typology Dismantler with identity meter · 3d
F2.2	Figure and ground; positive form and negative interval	O	GRM.f	I	Figure-Ground lab · img
F2.3	Visual weight and tension	O	GRM.f	I	Visual Gravity game
F2.4	Typology theory: archetypes, types, models	P	GRM.f	L, C	img
F2.5	Proportion and modular systems: classical, Renaissance, modular	P	GRM.f	C, I	Grid Lab · drw
F2.6	Islamic geometry as a generative system	P	GRM.f	C, I	Pattern Generator (star-and-polygon families) · drw
F2.7	East Asian modules: tatami and ken, the Yingzao Fashi	P	GRM.f	C	drw
F2.8	Textile grids: Andean and West African weaving systems	P	GRM.f	C	img
F2.9	Negative space and ma	P	GRM.f, GRM.s	C, I	Figure-Ground lab, spatial mode · img
F2.10	Structural skeletons and generative grids	P	GRM.f	I	Grid Lab: generate hierarchy from a module
F2.11	Scale and its cues	P	GRM.f	I	Scale Morph (object, room, block) · 3d
F2.12	Asymmetry, rhythm, controlled variation	P	GRM.f	I	Visual Gravity, series mode
F2.13	Typological subversion study, built parametrically	M	GRM.f, ANC	I, T2, D	Typology Dismantler authoring mode
F2.14	Form across 2D, 3D and digital: what changes, what holds	M	GRM.f	I	Scale Morph, medium mode
F2.15	Regional slot: a local form system read as generative	M	GRM.f	D	—

F3 Body and senses (13 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
F3.1	Affordance: how things say what to do	O	GRM.b	I	Guess-and-Reveal · img
F3.2	Colour interaction: the same colour is never the same	O	GRM.c	I	Colour Interaction Lab
F3.3	The body as metric: anthropometrics, reach, grip, posture	P	GRM.sn	L, I	Fit Tester · dat
F3.4	Cognitive load and interface constraints	P	GRM.i	L, I	Load Meter
F3.5	Joints, friction, movement	P	GRM.b	C	vid
F3.6	Colour physics: chroma, contrast, temperature, metamerism	P	GRM.c	L, I	Colour Interaction Lab, physics mode
F3.7	Colour across cultures	P	GRM.c	C, I	Colour Lab cultural overlays · img
F3.8	Light as hierarchy and depth	P	GRM.a	I	Light Sandbox · 3d
F3.9	Micro-interactions: click, resistance, weight, acoustic and haptic feedback	P	GRM.i	C, I	Micro-Interaction Detailer · aud
F3.10	Thresholds, sequence, duration	P	GRM.t, GRM.s	I	Threshold Sequencer
F3.11	Proxemics: distance as meaning	P	GRM.b	L, C	img
F3.12	Sensory system spec for your world	M	GRM.sn, .c, .a, .i, .t	S, D	Sense Engine (compound, §9)
F3.13	Regional slot: colour, light and distance where you live	M	GRM.c, .a	D	—

F4 Material and meaning (14 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
F4.1	Material as text	O	GRM.m	I	Material Card Deck · img
F4.2	Sign-swap: change the material, change the meaning	O	GRM.m	I	Sign-Swap · img
F4.3	Material behaviour: tensile, drape, thermal mass, algorithmic	P	GRM.m	L, I	Property Explorer

Unit	Concept	Depth	Field	Delivery	Interactive · media
F4.4	Provenance, sourcing ethics, lifecycle	P	GRM.m, PLW	L, C	dat
F4.5	Surface grammar: finish, reflectivity, friction, eye speed	P	GRM.m	I	Surface Lab · 3d
F4.6	Patina and designed decay	P	GRM.m, GRM.t	I	Patina Time Machine · img
F4.7	Semiotics: codes, metaphors, memory	P	GRM.n	L, C	img
F4.8	Language as material: typography principles	P	GRM.v	L, I	Type Lab · fonts
F4.9	Scripts of the world: Latin, Arabic, CJK, Devanagari and others	P	GRM.v	C, I	Script Pairing tool · fonts
F4.10	Tone of voice	P	GRM.v	I, P	Voice Tester with the Mirror
F4.11	Text in space: language as structural component	P	GRM.v, GRM.s	C	img
F4.12	Material dossier: five materials, lifecycle, meaning, ageing	M	GRM.m, PLW	D, S	Material Card Deck authoring; Planetary Ledger
F4.13	Multi-script text architecture	M	GRM.v	D, I	Script Pairing, system mode
F4.14	Regional slot: local materials and scripts	M	GRM.m, .v	D	—

F5 Systems and consequences (11 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
F5.1	Ecosystem map: what your intervention touches	O	PER	I	Ecosystem Map Builder
F5.2	Futures wheel	O	PER	I	Futures Wheel
F5.3	Complexity, feedback, stocks and flows	P	PER	L, C	ani, dat
F5.4	Non-human actors: algorithms, machines, agents	P	PER	L, C	Non-Human Actor simulator
F5.5	Climate, animals, land as actors	P	PER	C	dat
F5.6	Lifecycle, attrition, ecological weight	P	PLW	L, I	Planetary Ledger, level 1
F5.7	Scenario methods	P	PER	L, I	Scenario Generator
F5.8	What design cannot fix	P	PER	L, C	img

Unit	Concept	Depth	Field	Delivery	Interactive · media
F5.9	Scenario set with data for your intervention	M	PER	D, S	Scenario Generator with datasets · dat
F5.10	First planetary ledger	M	PLW	S	Planetary Ledger, level 2
F5.11	Regional slot: local systems, climate, water	M	PER, PLW	D	dat

Level 2 Method

Method is 58 units in four modules, taken as one continuous project; almost every unit is an interactive because the method is a set of moves, not a body of facts, and the lecture material is limited to naming the moves and showing one worked world.

M1 Research and premise (16 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
M1.1	From piles to patterns	O	PRM	I	Synthesis Board
M1.2	The perimeter: enable, protect, resist, reject, transform	O	PER	I	Perimeter Card
M1.3	Research routes: looking, asking, reading, archives, imagining	P	SRC	L, C	worked case · img
M1.4	Credibility, bias, the limits of evidence	P	SRC	L, I	Source Grader
M1.5	Research ethics and consent	P	SRC	L, D	templates
M1.6	Research questions and creative territories	P	PRM	I	Territory Mapper
M1.7	Clustering and pattern recognition	P	PRM	I	Synthesis Board, cluster mode
M1.8	The insight threshold: a claim you can act on	P	PRM	I, P	Insight Generator; Mirror voice
M1.9	Stance, worldview, productive contradiction	P	PRM	I	Contradiction Generator
M1.10	Human and experiential intention: scenarios, needs, resonance	P	PRM	I	Scenario Cards
M1.11	Narrative and semiotic architecture: stakes, archetypes, rituals	P	GRM.n	L, C, I	Archetype and Ritual Builder · img

Unit	Concept	Depth	Field	Delivery	Interactive · media
M1.12	Non-goals and strategic exclusions	P	PER	I	Perimeter Card, exclusion mode
M1.13	Hypothesis log	P	VER	I	Hypothesis Log
M1.14	Field research sprint: observation, interviews, artifacts	M	SRC, PRM	K, D, P	Historian voice · learner's own material
M1.15	Archive work: catalogues, primary texts, collections APIs	M	SRC	I, D	archive walkthroughs
M1.16	The defended premise	M	PRM	P, D	Client voice questions the premise

M2 Concept DNA (20 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
M2.1	Build a DNA for a supplied world	O	RUL	I	DNA Builder on a seed world
M2.2	The four tests: belongs, edge, failure, seductive-but-off	O	TST	I	Boundary Test Arena with cohort overlay · img
M2.3	The core operating system: proposition and structural tension	P	PRM, RUL	I	DNA Builder
M2.4	Constants, variables, thresholds	P	RUL	I	DNA Builder, threshold sliders
M2.5	Anti-rules and anti-DNA	P	ANT	I	DNA Builder
M2.6	The twelve grammars and choosing among them	P	GRM.*	L, I	Grammar Coverage Radar
M2.7	Formal and material grammars	P	GRM.f, .m	I	DNA Builder, linked to F2 and F4 outputs
M2.8	Chromatic, spatial, atmospheric, sensory grammars	P	GRM.c, .s, .a, .sn	I	DNA Builder, linked to F3
M2.9	Behavioural, interaction, temporal grammars	P	GRM.b, .i, .t	I	DNA Builder
M2.10	Narrative and verbal grammars	P	GRM.n, .v	I	DNA Builder, linked to M1.11, F4.10
M2.11	Construction grammar: how things are put together	P	GRM.k	I, C	DNA Builder · drw
M2.12	Alignment: practice, collection and masterbrand DNA	P	ALN	I	Alignment Map

Unit	Concept	Depth	Field	Delivery	Interactive · media
M2.13	Category obligations and controlled variation	P	OBL	I	Obligation Matrix
M2.14	Authoring the test set	P	TST	I	Boundary Test Arena, authoring mode
M2.15	The Stranger test: blind interpretation	P	BRF, VER	S, P	Stranger Lab (compound, §9)
M2.16	Versioning and the change log	P	VER	I	DNA Diff Viewer
M2.17	Full grammar authoring, all twelve, with exemplars	M	GRM.*	I, D	DNA Builder, full mode
M2.18	Three blind-interpretation rounds: peer, agent, both	M	BRF, TST	S, P	Stranger Lab; peer exchange
M2.19	Contributor brief and agent prompt architecture	M	BRF	I, T3, D	Brief Export; Prompt Architecture Studio
M2.20	DNA v1.0 release and defence	M	VER	P, D	Client and Stranger voices

M3 World architecture (10 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
M3.1	Roles in a collection	O	COL	I	Collection Matrix
M3.2	Architectural thesis and narrative arc	P	COL	I	Arc Planner
M3.3	Anchor, hero, core, entry, experimental, supporting	P	COL	I, C	Collection Matrix · img
M3.4	Families, sub-families, series, editions, programs, drops	P	COL	I	Collection Matrix, family mode
M3.5	Cross-scale coherence without cloning	P	COL	I	Coherence Checker
M3.6	Complexity, accessibility, cost and risk distribution	P	COL	I	Matrix overlays
M3.7	Pacing, curation, editing, rejection	P	COL	I	Cutting Room
M3.8	Carriers and interfaces: where the world meets people	P	COL	I	Carrier Map
M3.9	Full collection map with distribution	M	COL	S, D	Collection Matrix, full mode
M3.10	Editorial calendar and carrier strategy	M	COL	I, D	Drop Sequencer

M4 Make and test (12 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
M4.1	The prototype as a question	O	ANC	I	Prototype Question Planner
M4.2	Choosing practice and anchor	P	ANC	I	Anchor Chooser
M4.3	The adjacent translation	P	ADJ	I, P	Adjacent Translation Challenge; Stranger voice
M4.4	Technical translation and construction logic	P	GRM.k	L, C	drw
M4.5	Contextualisation: scenarios, rituals, use	P	PRM	I	Scenario Cards
M4.6	Prototyping and simulation fidelity; constraint negotiation	P	ANC	L, K	learner's own material
M4.7	Iterative critique, abandoned directions, revision history	P	VER, DOS	I, P	Abandoned Directions Archive; Mirror voice
M4.8	Specialist gaps	P	HND	I, P	Specialist Gap Finder; Fabricator voice
M4.9	The anchor expression, built	M	ANC	K, S	the chosen area's compound engine
M4.10	The differentiated family	M	ANC, COL	S	same
M4.11	The adjacent translation, built	M	ADJ	K, S	a second area's engine
M4.12	Evidence compilation and DNA v2.0	M	VER, DOS	D, P	Revision Timeline; all voices

Level 3 areas — S1 Space, S2 Object, S3 Body

Each area is one Orientation unit, a Practice lens of seven to nine units, and a Mastery studio that carries every unit of the original syllabus's studio at working depth, run on a free site, object or body pack with the partner as fabricator, client and stranger.

S1 Space and experience (34 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
S1.1	What a space makes you do	O	GRM.s	I	Read-a-Room, space mode · img

Unit	Concept	Depth	Field	Delivery	Interactive · media
S1.2	Program before form: adjacency, friction, privacy, zoning	P	GRM.s	I	Adjacency Builder
S1.3	Choreography: approach, threshold, compression, reveal	P	GRM.s, t	I	Sequence Choreographer
S1.4	Atmosphere: light, thermal, acoustic, olfactory	P	GRM.a	I	Atmosphere Sandbox, lite · 3d
S1.5	Reuse and palimpsest	P	GRM.s	I, C	Palimpsest Tool · img, drw
S1.6	The invisible: back of house, egress, services, money	P	HND	I	BOH Puzzle; Fit-Out Calculator
S1.7	Plan and section as a language	P	GRM.k	I, T1	Drawing Reader · drw
S1.8	Briefing a spatial designer	P	CMB	I, P	Brief Builder; Client voice
S1.9	Phenomenology of space: felt, not seen	M	GRM.a	L, I	Light Sandbox as the reading's lab
S1.10	Lineages of typology and form; regional modernisms	M	GRM.s	C	Courtyard-to-Tower morph · img, drw
S1.11	Typological subversion and hybrid programs	M	GRM.s, OBL	S	Typology Hybridiser
S1.12	Brand scenography: the DNA in three dimensions	M	ALN, GRM.s	S	Brand-to-Space Translator
S1.13	Circulation and behavioural framing	M	GRM.b, .s	S	World Engine, circulation layer (§9)
S1.14	The spatial palimpsest: keep, strip, overwrite	M	GRM.s	S, C	Palimpsest Tool, decision mode
S1.15	Heritage versus hostile intervention; structural audit and retrofit	M	HND	L, C, I	Retrofit Risk Checklist · drw
S1.16	Enclosure, joints, reveals, tolerances	M	GRM.k	I, T1	Detail Anatomy Explorer · drw
S1.17	Computational tectonics and fabrication	M	GRM.k	T2, T10	Node Playground; Toolpath Previewer
S1.18	Ageing, patina, maintenance cycles	M	GRM.m, LCY	I	Patina and Maintenance Timeline
S1.19	Climatic vernacular and thermal mass	M	GRM.a	S, T9	World Engine, climate layer · dat
S1.20	Motivated lighting, day to night	M	GRM.a	S	World Engine, light layer
S1.21	Olfactory and haptic architecture	M	GRM.sn	C	img

Unit	Concept	Depth	Field	Delivery	Interactive · media
S1.22	Acoustic character and soundscape	M	GRM.a	S	World Engine, acoustic layer · aud
S1.23	Edge condition and street activation	M	GRM.s	I	Edge Condition Configurator
S1.24	Master-planning and contextual adjacency	M	GRM.s	I	Context Massing · dat (OSM)
S1.25	Zoning, permissions, civic navigation	M	HND	L, C	—
S1.26	The back-of-house ecosystem	M	HND	S	World Engine, BOH layer
S1.27	Egress and life safety	M	HND	S	World Engine, egress check
S1.28	Services integration: mechanical, electrical, plumbing	M	HND	S	Ceiling Coordination puzzle
S1.29	CapEx, OpEx, yield, floor-area logic	M	RGT	S	World Engine, economics layer
S1.30	Tenancy, base-build, lease line	M	HND	I	Lease-Line Overlay
S1.31	Digital twin, walkthrough, clash detection	M	HND	T4	Walkthrough Template · 3d
S1.32	Schedules and handover	M	HND	I, D	Schedule Builder; Handover Auditor
S1.33	Studio brief: the site pack project, anchor and family	M	ANC, SITE	S, K, P	World Engine, full; Fabricator, Client, Stranger · pack
S1.34	Regional slot: a local site, a local climate	M	SITE	D	pack

S2 Object and product (40 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
S2.1	What an object makes you do	O	GRM.b	I	Read-a-Room, object mode · img
S2.2	The object as agent	P	GRM.b	C	img
S2.3	Hand and body: fit	P	GRM.sn	I	Fit Tester
S2.4	Use choreography and states	P	GRM.b, .t	I	Use Choreography Scriptor
S2.5	Cardboard prototyping	P	ANC	K	learner's own
S2.6	Making routes: craft to industry	P	GRM.k	I	Process Selector

Unit	Concept	Depth	Field	Delivery	Interactive · media
S2.7	The bill of materials, the box, the ship	P	HND, RGT	I	BOM Calculator; Flat-Pack Optimiser
S2.8	Repair and end of life	P	LCY	I	Disassembly Map
S2.9	Briefing a product designer	P	CMB	I, P	Brief Builder; Client voice
S2.10	Critical and speculative design	M	PRM	L, C	img
S2.11	Craft lineages and non-extractive translation	M	SRC, GRM.k	C, I	Craft-to-Industry Ladder · img
S2.12	Emotional durability	M	GRM.n	C	img
S2.13	Somatic mapping and anthropometrics	M	GRM.sn	S	Object Line, ergonomics layer (§9)
S2.14	Micro-interactions and haptic feedback	M	GRM.i	I	Micro-Interaction Detailer · aud
S2.15	Tactile temperature and material proximity	M	GRM.sn	I	Thermal Touch Chart
S2.16	Deconstructing archetypes	M	GRM.f	S	Typology Dismantler, object mode · 3d
S2.17	Spatial proxemics and object gravity	M	GRM.s	I	Room Gravity Planner
S2.18	State changes and kinetic volatility	M	GRM.t, .k	I	Kinetic State Explorer
S2.19	The physical buck and volumetric sketching	M	ANC	K	learner's own
S2.20	Digital-to-analog feedback loops	M	ANC	T2, T10	—
S2.21	The ergonomic prove-out	M	GRM.sn	K, I	Prove-Out Recorder
S2.22	Engineering comfort: foams, webbing, springs	M	GRM.k	S	Comfort Stack Simulator
S2.23	Pattern drafting for 3D volume	M	GRM.k	I	Upholstery Unfolder
S2.24	Hard-to-soft transitions	M	GRM.k	C	drw
S2.25	Mass manufacturing versus singular craft	M	GRM.k, RGT	S	Process Selector with cost curves
S2.26	Micro-tectonics, joints, tolerances	M	GRM.k	I	Tolerance Stack Visualiser
S2.27	Structural honesty	M	GRM.m	I	Material Truth Audit
S2.28	Load testing and fatigue	M	HND	S	Virtual Load Test
S2.29	Certification, fire and toxicity codes	M	HND	I	Compliance Path Finder
S2.30	IoT and hidden architecture	M	GRM.k	I, T8	Embedded Component Fit

Unit	Concept	Depth	Field	Delivery	Interactive · media
S2.31	Physical UI and digital state changes	M	GRM.i	I	State-Indicator Designer
S2.32	Algorithmic and parametric form generation	M	GRM.f	T2	Family Generator
S2.33	Flat-pack logic and volumetric efficiency	M	HND, PLW	S	Object Line, logistics layer
S2.34	BOM and value engineering against the DNA	M	RGT	S	Object Line, cost layer with DNA flags
S2.35	Packaging as the first threshold	M	GRM.t, .n	I	Unboxing Storyboard
S2.36	Right to repair and modular disassembly	M	LCY	I	Disassembly Map, full
S2.37	Patina and the aesthetics of ageing	M	GRM.m	I	Patina Time Machine
S2.38	End-of-life architecture	M	LCY, PLW	I	End-of-Life Router; Planetary Ledger
S2.39	Studio brief: the object family	M	ANC, SITE	S, K, P	Object Line, full; all voices · pack
S2.40	Regional slot: a local craft, a local supply chain	M	SITE	D	pack

S3 Body and wearables (46 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
S3.1	What a garment makes you do	O	GRM.b	I	Read-a-Room, garment mode · img
S3.2	Silhouette and power	P	GRM.f	I, C	Silhouette Timeline · img
S3.3	The body in motion and ease	P	GRM.sn	I	Ease Calculator
S3.4	Textile logic: grain, bias, knit	P	GRM.m	I	Grain and Bias Simulator
S3.5	Pattern basics	P	GRM.k	I, K	Pattern Unfolder · paper
S3.6	The semiotics of dress	P	GRM.n	I	Dress-Code Reader
S3.7	The collection matrix	P	COL	I	Range Plan Matrix
S3.8	The tech-pack	P	HND	I	Tech-Pack Builder
S3.9	Labour and circularity	P	PLW, LCY	I	Labour Ledger
S3.10	Briefing a fashion designer	P	CMB	I, P	Brief Builder; Client voice

Unit	Concept	Depth	Field	Delivery	Interactive · media
S3.11	Historiography of silhouette and power	M	SRC	C	img
S3.12	Avant-garde lineages and deconstruction	M	GRM.f, .k	C, I	Deconstruction Move Library · img
S3.13	Subcultural vernacular and trickle-up	M	GRM.n	C	img
S3.14	World dress histories and the textile trade	M	SRC	C	Route Map, textile layer · img
S3.15	Modesty as architecture	M	GRM.f, .s	C, I	Modesty Volume Studio
S3.16	The kinetic envelope and structural draping	M	GRM.k	K, S	Drape Simulator
S3.17	Somatic mapping and ease	M	GRM.sn	S	Capsule, body layer (§9)
S3.18	Volume, proportion, silhouette manipulation	M	GRM.f	I	Silhouette Distorter · 3d
S3.19	Woven versus knit; the bias	M	GRM.m	S	Grain and Bias Simulator, full
S3.20	Bio-fabrication and grown materials	M	GRM.m, PLW	C	img
S3.21	Embellishment as structure	M	GRM.k	I	Embellishment Explorer
S3.22	Drafting versus draping	M	GRM.k	K, I	Pattern Unfolder; half-scale form or paper
S3.23	Construction maps and seam architecture	M	GRM.k	I	Seam Architecture Selector · drw
S3.24	Closures, hardware, mechanical fastening	M	GRM.k, .i	I	Closure Engineering Picker
S3.25	Smart textiles and e-textiles	M	GRM.i	I, T8	E-Textile Circuit Sketcher
S3.26	Prosthetics and somatic augmentation	M	GRM.sn	C	img
S3.27	Thermal and microclimatic garments	M	GRM.a	S, T9	Heat-Stress Garment Simulator
S3.28	Clothing as armour and vulnerability	M	GRM.n	C	img
S3.29	Cultural coding, gender, subversion	M	GRM.n	I	Dress-Code Reader, full
S3.30	Adaptive architecture and sensory integration	M	GRM.i, .k	I	Adaptive Feature Picker

Unit	Concept	Depth	Field	Delivery	Interactive · media
S3.31	Grading logic and scale disjunction	M	HND	I	Grading Visualiser
S3.32	Hero, core, carry-over	M	COL	I	Range Plan Matrix with margins
S3.33	Range planning and drop sequencing	M	COL	I	Drop Sequencer
S3.34	Parametric patterning and 3D simulation	M	GRM.k	T4	Drape Simulator; a free simulation tool or CLO3D if licensed
S3.35	Digital fashion and post-physical wearables	M	COL	C, I	Digital Skin Economics
S3.36	Styling as generative design	M	GRM.n	I	Styling Variation Lab
S3.37	Art direction and the campaign lens	M	VEN	I	Campaign Brief Builder
S3.38	The runway as installation	M	GRM.t, .s	I	Runway Choreographer
S3.39	Technical flats and construction notation	M	HND	T1	Orthographic Trainer, garment mode
S3.40	The tech-pack as legal document	M	HND	I	Tech-Pack Error Hunt
S3.41	Tolerance, shrinkage, quality control	M	HND	I	Shrinkage Calculator
S3.42	The labour supply chain	M	PLW	I, C	Labour Ledger, full · dat
S3.43	Regenerative agriculture and sourcing	M	PLW	C	dat
S3.44	End-of-use and the circular economy	M	LCY	I	Mono-Material Checker
S3.45	Studio brief: the capsule	M	ANC, SITE	S, K, P	Capsule, full; all voices · pack (the learner's body)
S3.46	Regional slot: a local dress system as technology	M	SITE	D	img

Level 3 areas — S4 Art, S5 Identity, S6 Digital

S4 carries the original installation studio in full; S5 and S6 are new and written to the same depth, because they are the two practices the briefer lane commissions most.

S4 Art and installation (37 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
S4.1	What a room does to a work	O	GRM.s	I	Container Neutraliser · img
S4.2	The container: white cube, black box, street, desert	P	GRM.s	C	img
S4.3	From data to space	P	GRM.f	I	Translation Engine · dat
S4.4	Scale and site	P	GRM.s	I	Scale Power Slider · 3d
S4.5	Light and sound as material	P	GRM.a	I	Light and Sound Sandbox, lite
S4.6	Audience and time	P	GRM.b, .t	I	Audience Ritual Builder
S4.7	Documentation as the work	P	DOS	I	Archive-as-Artwork template
S4.8	Funding without selling out	P	RGT	P	Sponsor Negotiation, Client voice
S4.9	Commissioning an artist	P	CMB	I, P	Brief Builder
S4.10	Ideology of the container: institutional neutralisation	M	GRM.s	L, C	img
S4.11	Lineages: land art, light and space, situationists, institutional critique	M	SRC	C	img
S4.12	Relational and participatory theory	M	GRM.b	L, C	img
S4.13	Mythologies and the spatial sublime, without pastiche	M	GRM.n	C	img
S4.14	Translational methodologies: non-spatial data to physical form	M	GRM.f	S	Translation Engine, full
S4.15	Inductive versus deductive ideation	M	PRM	I	Ideation Router
S4.16	The what-if engine	M	PRM	I	Constraint Generator
S4.17	Site-specific versus site-responsive; the site as collaborator	M	SITE	C, I	Site Reading Checklist · pack
S4.18	Monumentality and the engineering of awe	M	GRM.s	S	Site Work, scale layer (§9)
S4.19	Micro-intervention and intimate proximity	M	GRM.s	S	Site Work, scale layer
S4.20	Light as mass and boundary	M	GRM.a	S	Site Work, light layer · 3d
S4.21	Ganzfeld environments and optical deprivation	M	GRM.a	S	Ganzfeld Simulator

Unit	Concept	Depth	Field	Delivery	Interactive · media
S4.22	Semantic materiality and the altered artifact	M	GRM.m	C	img
S4.23	Organic volatility and implied threat	M	GRM.t, .m	S	Volatility Timeline
S4.24	Frequencies as spatial boundaries; infrasound	M	GRM.a	S	Site Work, sound layer · aud
S4.25	Acoustic isolation and the vacuum	M	GRM.a	S	Anechoic versus Reverberant
S4.26	The site-specific soundscape	M	GRM.a	K, I	field recording; Soundscape Mixer · aud
S4.27	Spatial computing and XR interventions	M	GRM.s	T4	AR Overlay Template
S4.28	The algorithm as co-author	M	GRM.t	S	Live-Data Installation Sim · dat
S4.29	Post-human spectatorship	M	GRM.b	C, I	Non-Human Actor simulator
S4.30	The complicit spectator: rules of engagement	M	GRM.b	I	Audience Ritual Builder, full
S4.31	Durational engagement and ephemerality	M	GRM.t, LCY	I	Ephemerality Planner
S4.32	The archive as artwork	M	DOS	I	Archive template, full
S4.33	Spatial occupation and the right to the city	M	PER	C	img
S4.34	The ecological debt of the ephemeral	M	PLW	S	Ephemeral Debt Calculator; Planetary Ledger
S4.35	Patronage, grants, brand commissions	M	RGT	P	Sponsor Negotiation, full
S4.36	Studio brief: the site proposal	M	ANC, SITE	S, K, P	Site Work, full; all voices · pack
S4.37	Regional slot: a local site, a local mythology	M	SITE	D	pack

S5 Identity and communication (24 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
S5.1	What a mark makes you feel	O	GRM.n	I	Mark Reader · img
S5.2	The brand as a world	P	ALN	C	img
S5.3	Marks: what a symbol can carry	P	GRM.f	I	Mark Lab

Unit	Concept	Depth	Field	Delivery	Interactive · media
S5.4	Type as a system	P	GRM.v	I	Type System Builder · fonts
S5.5	Colour as a system	P	GRM.c	I	Colour System Builder
S5.6	Tone of voice	P	GRM.v	I, P	Voice Tester
S5.7	Grids and layout	P	GRM.f	I	Grid Lab, layout mode
S5.8	Multi-script and bilingual systems	P	GRM.v	I	Script Pairing, system mode
S5.9	Campaign as world extension	P	VEN	I	Campaign-to-Premise Checker
S5.10	Writing the design brief	P	CMB	I, P	Brief Builder
S5.11	Identity histories: railways, airlines, institutions, cities, startups	M	SRC	C	img
S5.12	Semiotics of the mark	M	GRM.n	L, C	img
S5.13	Logo construction and geometry	M	GRM.f	I, T2	Mark Lab, construction mode
S5.14	Typographic systems and hierarchy	M	GRM.v	I	Type System Builder, full
S5.15	Type across scripts: matching, mirroring, rhythm	M	GRM.v	I	Script Pairing, full
S5.16	Colour systems and accessibility	M	GRM.c	I	Colour System with contrast checks
S5.17	Layout systems and responsive grids	M	GRM.f	I	Grid Lab, responsive mode
S5.18	Motion and sound identity	M	GRM.t, .a	T4	Motion Lab · aud
S5.19	Voice and verbal grammar in depth	M	GRM.v	I, P	Voice Tester, full
S5.20	Touchpoint architecture	M	VEN	I	Touchpoint Stress Test
S5.21	Guidelines a stranger can apply	M	BRF	S	Stranger Lab, identity mode
S5.22	Campaign architecture	M	VEN	I	Campaign Builder
S5.23	Studio brief: the identity system	M	ANC, SITE	S, P	Identity Engine, full (§9) · pack (a public-domain identity to re-world)
S5.24	Regional slot: a multi-script identity for a local institution	M	SITE	D	fonts

S6 Digital product and interface (20 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
S6.1	What a screen makes you do	O	GRM.i	I	Read-a-Room, screen mode
S6.2	Interface as affordance	P	GRM.i	C, I	Affordance Guess-and-Reveal, screen mode
S6.3	Flows, states and errors	P	GRM.i, .t	I	Flow Mapper
S6.4	Design systems as DNA	P	GRM.*, BRF	I	Design Tokens from DNA
S6.5	Content and microcopy	P	GRM.v	I, P	Copy Lab
S6.6	Accessibility	P	OBL	I	Accessibility Checker
S6.7	Prototyping	P	ANC	T4	Interface Prototype Kit
S6.8	Writing the product brief	P	CMB	I, P	Brief Builder
S6.9	Histories of the interface	M	SRC	C	img, vid
S6.10	Information architecture	M	GRM.s	I	IA Builder
S6.11	Interaction patterns and their affordances	M	GRM.i	I	Pattern Library
S6.12	State machines and error design	M	GRM.i, .t	I	State Machine
S6.13	Tokens, components, variants	M	GRM.*, BRF	I	Design System Builder
S6.14	Motion and feedback	M	GRM.t, .i	T4	Motion Lab
S6.15	Content design and voice	M	GRM.v	I, P	Copy Lab, full
S6.16	Accessibility to WCAG	M	OBL	I	Accessibility Audit
S6.17	Data, privacy, sovereignty	M	PER	L, T3	Ethics Review checklist
S6.18	From prototype to handover spec	M	HND	I, S	Product Loop (§9); Handover Auditor
S6.19	Studio brief: the product	M	ANC, SITE	S, P	Product Loop, full · pack (a well-known flow to re-world)
S6.20	Regional slot: localisation and scripts	M	SITE	D	fonts

Tools T1–T10

Every tool concept in the original can grow along one path — a single lab, then a coupled lab that takes another unit's output as its input, then a layer in one of the compound engines of §9 — and the tables below give each tool its units and that path; T1–T5 are the core thread at Practice, T6–T10 are Mastery electives, and every tool has a free route.

Evolution study (the same tool, three sizes)

Tool concept	First used in	Single lab	Coupled lab	Engine layer	Free route
T1 Seeing and drawing	F2, S1.7	Orthographic Trainer: draw the plan of a rotating object with overlay feedback	Draw the section of your own Adjacency Builder plan; the trainer checks it against the model	World Engine and Object Line accept the learner's drawings as inputs, and the Handover Auditor reads them	pencil, paper, phone camera; in-app canvas
T2 Parametric thinking	F2.13, M3	One rule, many outputs in p5.js	The Typology Dismantler's sliders are exposed as a node graph the learner rewires	Family Generator drives the Collection Matrix: change a rule, the whole family and its costs update	p5.js in the browser; Blender geometry nodes; Grasshopper only if licensed
T3 Working with AI	M2.15	Prompt Architecture Studio with drift score	Agent Team Builder briefs three roles from the DNA and compares their outputs	Stranger Lab across media (text, image, 3D sketch); the partner's voices are built on this	free model tiers or program-provided access, logged

Tool concept	First used in	Single lab	Coupled lab	Engine layer	Free route
T4 Simulation	F3.8, S1.31	A scene, a light, a walk (three.js template)	The Sequence Choreographer's path becomes the walk; the Light Sandbox's lights are the scene's lights	World Engine, Site Work and Product Loop are real-time engines with layers switched on per unit	three.js templates; Godot for heavier work
T5 Honest evidence	M4, S1.31	Render Honesty Audit: what does this picture claim about scale, material, cost	PBR Material Lab fed by the Material Card Deck's chosen materials	Every engine's export runs through the audit before it enters the dossier	Blender
T6 Reality capture	S1.5, S2.19	Phone photogrammetry of an object with a quality checker	The capture becomes the buck in the Object Line, or the existing building in the Palimpsest Tool	World Engine on a captured site	free photogrammetry apps and open point-cloud viewers
T7 Data visualisation	M1, F5	Encoding Chooser on a supplied dataset	Synthesis Board data becomes a node map; Route Map takes the learner's data	Translation Engine (S4) spatialises data into the Site Work engine	d3 in-app; open datasets
T8 Physical computing	S2.30, S3.25	Sensor-to-actuator logic in a browser simulator	State-Indicator Designer runs on the simulated board	Object Line's cyber-physical layer	free microcontroller simulators; hardware optional
T9 Environmental analytics	F5.6, S1.19	Climate Simulation Lite for a latitude	Planetary Ledger takes the BOM from Object Line or the materials from the Material Dossier	World Engine's climate layer; Capsule's heat-stress layer; the final ledger	open climate tools; openLCA

Tool concept	First used in	Single lab	Coupled lab	Engine layer	Free route
T10 Fabrication	S1.17, S2.20	Toolpath Previewer on a supplied part	Print Failure Predictor on the learner's own Family Generator output	Object Line's making layer: route, tolerance, failure, cost together	free CAM previewers; slicer software

T1 Seeing and drawing (5 units)

Unit	Concept	Depth	Delivery	Interactive · media
T1.1	Observational and analytical drawing: gesture, line weight, mass, silhouette	P	K, I	Line-Weight Coach · learner's photos
T1.2	Orthographic logic: plan, section, elevation, exploded axonometric	P	I	Orthographic Trainer · 3d
T1.3	Movement notation and state changes	P	I	Movement Notation Sketcher · vid
T1.4	Ideation sketching and the physical buck	P	K	paper, card
T1.5	Full orthographic and exploded set of a real assembly	M	K, D	Orthographic Trainer, authoring · drw

T2 Parametric thinking (5 units)

Unit	Concept	Depth	Delivery	Interactive · media
T2.1	One rule, many outputs	P	I	p5.js sketchpad
T2.2	Topology: NURBS, meshes, edge flow, watertightness	M	L, I	Topology Lab · 3d
T2.3	Node-based logic and cascading change	M	I	Node Playground

Unit	Concept	Depth	Delivery	Interactive · media
T2.4	Creative coding and shader logic	M	I	Shader Sketchpad
T2.5	Form-finding and structural optimisation	M	S	Force Form-Finder

T3 Working with AI (5 units)

Unit	Concept	Depth	Delivery	Interactive · media
T3.1	Prompt architecture: constraints, references, negatives	P	I	Prompt Architecture Studio
T3.2	Agent briefs and agent teams	P	I	Agent Team Builder
T3.3	Multimodal synthesis constrained by the DNA; drift scoring	P	S	Drift Detector
T3.4	Workflow and folder architecture; versioning	M	I, D	Studio OS Kit
T3.5	Bias, data sovereignty, surveillance	M	L, I	Ethics Review checklist

T4 Simulation and interface (5 units)

Unit	Concept	Depth	Delivery	Interactive · media
T4.1	A scene, a light, a walk	P	I	three.js template · 3d
T4.2	Real-time engines and the digital twin	M	S	Walkthrough Template, full
T4.3	Simulation mechanics: collision, physics, triggers	M	I	Trigger Sandbox
T4.4	Interface prototyping and full-stack sketches	M	I	Interface Prototype Kit

Unit	Concept	Depth	Delivery	Interactive · media
T4.5	Sound in a DAW; spatial audio	M	I	Spatial Audio Mixer · aud

T5 Honest evidence (4 units)

Unit	Concept	Depth	Delivery	Interactive · media
T5.1	What a render claims	P	I	Render Honesty Audit · img
T5.2	Physically based materials	M	I	PBR Material Lab · 3d
T5.3	Cinematography, framing, motivated light	M	I	Cinematography Coach
T5.4	Post-production ethics	M	I, D	Ethics checklist

T6 Reality capture (3 units)

Unit	Concept	Depth	Delivery	Interactive · media
T6.1	Photogrammetry, radiance fields, splats	M	K, I	Capture Quality Checker · 3d
T6.2	LIDAR and point clouds	M	I	Point Cloud Explorer · open scans
T6.3	Metrology and micro-scanning	M	C	vid

T7 Data visualisation (4 units)

Unit	Concept	Depth	Delivery	Interactive · media
T7.1	Parsing, cleaning, sorting	M	I	Data Parser · dat
T7.2	Choosing the encoding	M	I	Encoding Chooser
T7.3	Dashboards and node maps	M	I	Node-Map Builder

Unit	Concept	Depth	Delivery	Interactive · media
T7.4	Spatialising data	M	S	Data-to-Space Mapper

T8 Physical computing (4 units)

Unit	Concept	Depth	Delivery	Interactive · media
T8.1	Microcontrollers and logic boards	M	I	browser board simulator
T8.2	Sensory input and biometric capture	M	I	Sensor Logic Builder
T8.3	Actuation and kinetic output	M	I	Behaviour State Machine
T8.4	A responsive behaviour, end to end	M	S, K	simulator or hardware

T9 Environmental analytics (2 units)

Unit	Concept	Depth	Delivery	Interactive · media
T9.1	Climatic simulation: sun, wind, thermal bridging	M	S	Climate Simulation · dat
T9.2	Life cycle assessment	M	S	Planetary Ledger, level 3 (openLCA)

T10 Fabrication (3 units)

Unit	Concept	Depth	Delivery	Interactive · media
T10.1	CAM logic, toolpaths, G-code	M	I	Toolpath Previewer
T10.2	Additive manufacturing constraints	M	I	Print Failure Predictor
T10.3	Failure analysis and material physics	M	S, C	Failure Cases; Virtual Load Test

Level 4 Venture and release R1–R5

Five modules and 52 units, all of the original's venture and release content with the duplication removed; every number-bearing unit is a simulator with a visible formula, and every legal unit is labelled literacy, not advice.

R1 The world as venture (12 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
R1.1	One world, many carriers	O	VEN	I	Orbital Architecture Mapper
R1.2	Masterbrand positioning and non-negotiable codes	P	VEN, ALN	I, C	Brand Code Stress Test · img
R1.3	Orbital architecture: endorsed sub-brands, ventures, acquisitions	P	VEN	I	Orbital Mapper, rules mode
R1.4	Ecosystem orchestration across verticals	P	VEN	I	Vertical Flow Map
R1.5	Exclusivity and access	P	VEN	I	Access Tier Designer
R1.6	Ritual and tribal signifiers	P	VEN, GRM.n	I	Ritual and Signifier Board
R1.7	Feedback loops and co-creation	P	VEN	I	Feedback Loop Architect
R1.8	Campaign logic: the campaign as world extension	P	VEN	I	Campaign-to-Premise Checker
R1.9	Bilingual and multi-script communication architecture	P	GRM.v	I	Script Pairing, campaign mode
R1.10	Activation mechanics: registration, tracking, journey	P	VEN	I	Activation Workflow Builder
R1.11	Full masterbrand and orbital architecture with rules of engagement	M	VEN	S, D	Orbital Mapper, full
R1.12	Campaign architecture	M	VEN	S, D	Campaign Builder

R2 Money, law and rights (11 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
R2.1	How long until you run out	O	RGT	I	Runway Simulator

Unit	Concept	Depth	Field	Delivery	Interactive · media
R2.2	Bootstrapping versus institutional capital; dilution	P	RGT	L, I	Dilution Calculator
R2.3	Patronage, grants, public-private partnership	P	RGT	L, C	—
R2.4	Cash flow, burn, receivables against deposits	P	RGT	S	Runway Simulator, full
R2.5	Intellectual property: marks, design rights, patents, copyright	P	RGT	I	IP Decision Tree
R2.6	Contracts, NDAs, transfer of rights	P	RGT	I	Contract Clause Explorer
R2.7	Cultural sovereignty and defensive protocols	P	RGT, SRC	L, I	Copy Response Playbook
R2.8	Attribution and consultation as protocol	P	SRC	I, D	template from F1.18
R2.9	18-month model with two scenarios	M	RGT	S, D	Venture Console (§9)
R2.10	IP and attribution protocol for the project	M	RGT	D	—
R2.11	Contract anatomy workshop (literacy)	M	RGT	L, I	Clause Explorer, full

R3 Making it real (12 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
R3.1	Who does what	O	HND	I	Collaboration Matrix
R3.2	The studio operating system: folders, versions, templates, agents	P	HND	I, T3	Studio OS Kit
R3.3	Pitch architecture and defence	P	DOS	I, P	Pitch Builder; Client voice
R3.4	Resource management and specialist orchestration	P	HND	I	Specialist Vetting Scorecard
R3.5	The collaboration matrix: dependencies, conflict, risk	P	HND	I	Collaboration Matrix, full
R3.6	Minimum viable implementation	P	HND	S, P	MVI Negotiator; Fabricator and CFO voices
R3.7	Release sequencing and phased drops	P	COL	I	Drop Sequencer
R3.8	The anatomy of the handover	P	HND	I	Handover Package Builder
R3.9	Governance, compliance, permissions	P	HND	I	Compliance Path Finder
R3.10	Briefing others: the brief as artifact	P	CMB	I, P	Brief Builder, full

Unit	Concept	Depth	Field	Delivery	Interactive · media
R3.11	A complete handover package, audited	M	HND	S, D	Handover Auditor
R3.12	MVI negotiation at scale	M	HND	S, P	Venture Console with the DNA constants

R4 Responsibility (8 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
R4.1	The life after launch	O	LCY	I	Lifecycle Planner
R4.2	Sourcing, substitution, budget prioritisation	P	LCY, RGT	I	Substitution Trade-Off
R4.3	Packaging, distribution, interface	P	LCY	I	Unboxing and Distribution Storyboard
R4.4	Operation, maintenance, repair	P	LCY	I	Maintenance Budget
R4.5	End-of-life architecture	P	LCY, PLW	I	End-of-Life Router
R4.6	The planetary ledger	P	PLW	S	Planetary Ledger, level 2
R4.7	The final ledger with data	M	PLW	S, D	Planetary Ledger, level 3
R4.8	Repair and end-of-life design for the studio project	M	LCY	S, D	area engine, lifecycle layer

R5 Defence and dossier (9 units)

Unit	Concept	Depth	Field	Delivery	Interactive · media
R5.1	A transfer brief in miniature	O	DOS	I	Transfer Brief Generator
R5.2	Systemic transfer: the method under an unfamiliar constraint	P	DOS	I	Transfer Brief Generator, full
R5.3	Critical defence: evidence, limits, the evolution of versions	P	DOS	L, P	all voices
R5.4	Dossier curation: alternatives, abandoned directions, history	P	DOS	I	Dossier Builder
R5.5	Specialist gap identification	P	DOS	I	Specialist Gap Finder
R5.6	Future-practice plan	P	DOS	I, D	Plan template

Unit	Concept	Depth	Field	Delivery	Interactive · media
R5.7	The transfer brief, executed	M	DOS	S, K	the chosen engine under a new constraint
R5.8	The defence room	M	DOS	S, P	Defence Room: three-voice panel
R5.9	The professional dossier	M	DOS	D	Dossier Builder, export

Compound simulations

Single labs teach one concept; the program's central claim — that a world is a system in which form, body, material, money and consequence move together — can only be taught by simulations that couple several concepts, so ten engines are defined here, each built as one model with layers that units switch on one at a time.

The principle. An engine holds one state (a site, an object, a garment, a DNA, a venture) and several coupled models. Early units switch on one layer and the engine behaves like a single lab. Later units switch on more, and the learner sees the couplings: change the wall thickness and the cost, weight, shipping volume, carbon and load safety all move; change a threshold in the DNA and the Stranger's outputs drift differently across media. Each engine's layers list is also its build order, so the first version is small.

Engine	State it holds	Layers (each added by a unit)	Units served	What it teaches that single labs cannot
1. Timeline Atlas	The world set on a map and a timeline	Dominant and obscured strands; trade routes with object flows; object autopsy; credit map; regional layer; authoring for the learner's own lineage	F1 all; S3.14; S1.10; S5.11; S6.9	That history is a network with flows, not a list; that any object sits at a crossing of routes, labour and meaning

Engine	State it holds	Layers (each added by a unit)	Units served	What it teaches that single labs cannot
2. Sense Engine	One scene with a body in it	Anthropometric fit; cognitive load; colour interaction with cultural overlays; light hierarchy; micro-interaction feedback; threshold sequence; proxemics	F3 all; F3.12	That colour changes with light, that light changes what the body does, that sequence changes both
3. Stranger Lab	The learner's DNA	Text outputs; image outputs; 3D sketches; peer interpretation; drift scoring per grammar; rule editing and rerun; version comparison	M2.15, M2.18, M4.3, S5.21, T3.3	That generativity is measurable; which rule is weak; how a fix in one grammar changes drift in another
4. World Engine (space)	A site pack: plan, context, climate	Adjacency and zoning; circulation and sequence with a tension curve; daylight and motivated light; thermal mass and shading; acoustic character; back of house and egress; services coordination; CapEx, OpEx and yield; the learner's own drawings as input	S1.13–S1.33; F5; T4; T9; R3	That a spatial decision is never only spatial: move a wall and the daylight, the staff path, the reverberation and the rent per square metre all move

Engine	State it holds	Layers (each added by a unit)	Units served	What it teaches that single labs cannot
5. Object Line	An object family from the Family Generator	Ergonomic fit; kinetic states; process route and tolerances; comfort stack for soft goods; load and fatigue; embedded components; BOM and value engineering with DNA flags; flat-pack and shipping; carbon and end of life; making constraints from fabrication	S2.13–S2.39; T2; T8; T9; T10; R4	That the box, the factory and the recycling stream are design decisions made at the sketch
6. Capsule	A body and a range	Ease and motion; grain, bias and drape; pattern and seams; grading across sizes; heat stress and microclimate; range plan with margins; tech-pack completeness; labour and material ledger	S3.16–S3.45; T4; T9	That a fabric choice changes drape, yield, size grade, factory spec, cost and labour at once
7. Site Work (installation)	A public site and a duration	Scale against the body; light as mass; sound as boundary; audience rules and journey; decay or de-install schedule; live-data co-authorship; ephemeral debt; sponsor negotiation	S4.14–S4.36	That awe, intimacy, complicity and ecological cost trade against one another in the same room

Engine	State it holds	Layers (each added by a unit)	Units served	What it teaches that single labs cannot
8. Identity Engine	The DNA's grammars	Mark; type system; colour system with contrast; tone of voice; grids and responsive layout; multi-script pairing; motion and sound; touchpoint stress test; guidelines tested by the Stranger	S5.13–S5.23; R1	That an identity is the DNA made visible, and that one weak grammar shows up on every touchpoint
9. Product Loop	A flow and a design system	Information architecture; states and errors; tokens and components from the DNA; copy and voice; accessibility audit; motion; handover spec	S6.10–S6.19	That accessibility, content and tokens are one system, not three checks
10. Venture Console	The world as a business	Runway and burn; dilution scenarios; margin stacks per carrier; IP routes and costs; MVI trade-offs against DNA constants; orbital architecture; handover readiness; the planetary ledger	R1–R4; S1.29; S2.34; S3.32	That cutting cost or time breaks named constants, and that a funding route has a price in control

Cross-engine couplings worth building. Object Line to Venture Console (BOM into margin); World Engine to Venture Console (fit-out into runway); every engine to the Planetary Ledger (materials and energy into carbon); every engine to the Stranger Lab (outputs judged against the DNA); every engine to the Handover Auditor (exports checked for what a maker would misread).

Build staging. Each engine starts with two coupled layers and a visible formula panel, adds one layer per unit as the content is written, and is never built past the units that exist. The Stranger Lab, Sense Engine and Timeline Atlas come first because Foundations and Method need them; World Engine and Object Line next because they prove the studios; the rest follow their areas.

Media assets study

About 70% of the units need media beyond text, and nearly all of it can be sourced free or produced with the tools in this account; the expensive kind of asset (commissioned photography, filmed practitioners) is replaced by open collections, public talks and drawings we make ourselves.

Asset type	Units that need it	Volume	Free source	Produce ourselves
Object and artwork images (img)	Every C unit; F1 world sets; S2, S3, S5 cases	40–60 per module, about 1,200 total	CC0 museum collections and their APIs; Wikimedia; public-domain surveys	none needed; licence note per image
Drawings, plans, sections (drw)	F2, S1, S2.24, S3.23, T1	150–250	Archnet; public-domain survey drawings; Wikimedia; OSM for plans	SVG drawings redrawn to one house style, in-app
3D models (3d)	F2.1, F3.8, S1, S2, S4, T4, T5	60–100	Smithsonian 3D; CC models on Sketchfab; Scan the World	Blender models for archetypes and site packs
Short video (vid)	F1.13, F3.5, S6.9, T1.3, T6.3; anything where motion carries meaning (drape, mechanisms, light changing)	40–60 clips, under 8 minutes	embedded public talks and museum channels	screen recordings of the engines; short AI-narrated explainers only where text fails

Asset type	Units that need it	Volume	Free source	Produce ourselves
Animated diagrams (ani)	F1.1, F5.3, O1.3, M2.6, and any process explained	80–120	—	SVG and JS animations in-app; Figma for stills
Audio (aud)	F3.9, S1.22, S2.14, S4.24–26, T4.5, S5.18	60–100 clips	CC field recordings; the learner's own	synthesised feedback sounds; spatialised demos
Data (dat)	F1.11, F5, S1.19, S1.24, T7, T9, R2	20–30 sets	Our World in Data; World Bank; UN; climate services; OSM	cleaned subsets with attribution
Fonts	F4.8–9, S5, R1.9	30–50 families	Google Fonts	none
Site, object, body, brand packs (pack)	S1.33, S2.39, S3.45, S4.36, S5.23, S6.19 and regional slots	2–3 per area	OSM plans; CC0 buildings; public scans; public-domain identities; well-known flows	assembled and documented per area
Case dossiers	Every C unit	about 300	the facts from open sources	one page each: story, image, licence, which units use it
Partner prompts	Every P unit; the five voices	60–80 prompt architectures	—	written, tested, versioned

Rules. A unit's media list is part of its spec; no unit is Drafted without its media sourced or scheduled. Images are referenced by link where licence requires and stored where CC0 allows. Every asset carries its licence, source and the units that use it, so a takedown or a swap touches one record.

Lecture material study

Lecture material is whatever must be told rather than discovered — sequenced history, argument, definitions, procedures, warnings — and the program keeps it to about a fifth of learner time by converting each kind into the form that lets the learner meet it through doing first.

What counts as lecture material

Kind	Examples in this program	Why it must be told
Sequenced narrative	F1's pivots and centres; S3.11 history of silhouette; S5.11 identity histories	Order and causation are the content
Theory and argument	S1.9 phenomenology; S4.12 relational theory; F2.4 typology theory; S2.10 critical design	The claim and the counter-claim are the content
Definitions and taxonomies	M2.6 the twelve grammars; M3.3 roles; R2.5 IP types	Shared vocabulary is needed before practice
Procedures	M1.3 research routes; R3.8 handover anatomy; M1.5 ethics and consent	Steps, in order, with the reasons
Cases	Every C unit	The specific is what makes the general stick
Warnings and ethics	F1.18 attribution; T3.5 bias; R2.7 sovereignty	Consequences must be named

How each kind converts

Kind	Conversion	What remains as reading or video
Sequenced narrative	Told through the Timeline Atlas and object stories; the learner navigates before reading	One 400-word card per pivot or centre: why it mattered, what to look for

Kind	Conversion	What remains as reading or video
Theory and argument	An argument card (claim, counter-claim, one case) followed by the lab that makes the theory felt: phenomenology through the Light Sandbox, relational theory through the Audience Ritual Builder	Short original essays, because the canonical texts are not free and a summary that replaces them is not allowed
Definitions and taxonomies	Sort first, then reveal: the learner sorts examples into bins, the taxonomy is named after	A glossary entry and a one-line "say it to a designer"
Procedures	Embedded as the interactive's steps; the tool is the procedure	A checklist to keep
Cases	World-set dossiers tagged to units; never lectured	The dossier page itself
Warnings and ethics	Decision simulators: the Spectrum Sort, the Sponsor Negotiation, the MVI Negotiator	A principle stated once, in a sentence

Format rules for what remains. Text first, with optional narrated audio; video only where motion carries meaning; nothing over eight minutes; every lecture element names the lab that precedes it and the artifact field it feeds; the original's voice is allowed in provocations and studio briefs, nowhere else.

Share of time, estimated across the comprehensive version. Lecture material about 20%; cases about 15%; single labs about 25%; compound simulations about 20%; making and documents about 15%; the partner about 5%. Foundations is heavier in lecture and cases; Method and the studios are heavier in labs and engines.

Research log and evolution rules

Research is uncapped; what is capped is losing it, so every module carries a research log, every finding that changes the structure is recorded as a version, and the structure itself expects to be wrong in places.

The research log per module (one living document each): questions open; sources found, with licence and quality grade; world-set candidates and the units they would serve; claims sheet (every factual statement in unit copy with its source); findings that argue for a structural change; open questions for you. Research sprints are dated entries in the log and can be re-run at any time.

How a finding changes the structure

Finding	Structural move	Version effect
A concept the units do not cover	Add a unit, at the depth the concept needs	minor (v2.x)
A unit that is really two	Split	minor
Two units that are one	Merge; keep both numbers as aliases	minor
A concept sitting at the wrong depth	Re-depth (O to P, P to M, or back)	minor; slices update automatically
A module that belongs elsewhere	Move; update the artifact fields it feeds	major (v3.0)
A better engine coupling	Add a layer to an engine; note the unit that switches it on	minor
A media asset that cannot be sourced free	Swap or produce; record in the asset register	none

Change log (append here)

- v2.1 — Pass 1 master structure created from Concept v2; 21 modules, 10 tools modules, 10 engines, about 430 units.
- v3.0 — Audit against all earlier versions (see Final Structure v3.0): 22 units added, 2 renamed, six threads, closing tests, prerequisites, seven partner voices, commissioning variants of every studio brief. Unit count about 450. The additions register below is appended here so this document stays the single unit register.

Additions register, v3.0 (columns as the unit tables; Follows = the unit each new one sits after)

Unit	Concept	Depth	Field	Delivery	Interactive · media	Follows
F1.3a	Interdisciplinary discourse: what film, performance, sound, gaming and digital culture teach design	P	SRC, GRM.n	L, C	Cross-Discipline Case Wall · vid, aud, img	F1.3
F1.20a	Vernacular intelligence: oral history, folktale and local taxonomy as sources, with consent	M	SRC	L, K, D	Oral History Kit · aud	F1.20
F4.7	Renamed: Semiotics, symbolism and collective memory	P	GRM.n	L, C	Sign-Swap · img	—
F5.5a	Land, water and climate as the site	P	PER, PLW	L, C, I	Site Systems Map · dat	F5.5
M1.11a	World-building lessons from film, games and performance	P	GRM.n, GRM.t	L, C	Bible Anatomy · vid	M1.11
S1.1a, S2.1a, S3.1a, S4.1a, S5.1a, S6.1a	The practice DNA of this medium: what it adds and forbids	P	ALN	L, I	Practice DNA card	Sx.1
S1.8a	Exhibition, museum and cultural-district experience	P	GRM.s, GRM.n	C, I	Visitor Journey Mapper · img, drw	S1.8
S1.8b	The service script behind the space: hospitality, food and drink, operations	P	GRM.b, HND	C, I	Service Blueprint Builder	S1.8a
S1.24a	The outdoor room: landscape, shade, water, ground	M	GRM.s, GRM.a	C, S	World Engine, outdoor layer · dat	S1.24
S3.35a	Accessories and jewellery: the object on the body	M	GRM.f, GRM.k	C, K, I	Fit Tester, body mode · img	S3.35
T4.2a	Data-rich models: BIM and PLM	M	HND	L, I	Data-Rich Model Explorer · 3d	T4.2
T5.1a	Documenting physical work: photography, capture, sequencing	P	DOS	L, K, I	Documentation Planner · img	T5.1
R1.10a	The physics of the checkout	P	VEN	C, I	Checkout Flow Dissector	R1.10

Unit	Concept	Depth	Field	Delivery	Interactive · media	Follows
R1.10b	Online-to-offline synchronisation	P	VEN	I	O2O Journey Mapper	R1.10a
R1.10c	The flagship as media	P	VEN	C, I	Media-Value Calculator · img	R1.10b
R1.12a	Large-scale activation: expos, festivals, civic events	M	VEN	C, S	Activation Workflow, scale mode · img	R1.12
R2.4a	Cost, value and margin architecture	P	RGT	L, S	Venture Console, margin layer	R2.4
R2.4b	Expansion across markets	P	RGT	L, I	Expansion Canvas	R2.4a
R2.6	Renamed: Contracts, NDAs, licensing, transfer of rights	P	RGT	I	Contract Clause Explorer	—
R3.6a	Client and stakeholder negotiation	P	HND, CMB	S, P	Negotiation Simulator; Client voice	R3.6
R4.4a	Lifecycle economics: total cost of ownership	P	LCY, RGT	S	Venture Console, lifecycle layer	R4.4

Rule changes. Every Sx studio brief has Make (ANC) and Commission (CMB) variants; T4 retitled Simulation, interface and data-rich models; T5 retitled Honest evidence and documentation; S1.21 gains scent and culinary case dossiers.

- v3.1 and v3.2 — Final Structure revisions with no unit changes: presentation forms and the consumption matrix (v3.1); no fixed unit sequence, ten opening moves and four invariants (v3.2).
- v3.3 — F1 redesigned in Pass 2: 23 units (21 plus two v3.0 additions) become 31; the pivots become recurrences, the continents become worlds joined by networks, eight units added; F1 renumbered with a Was mapping.
- v3.4 — F1 sprints 2 and 3: F1.19a War, crisis and making, F1.22a Use and consumption and F1.30a Counterfactual history added; 34 units.
- v3.5 — F1 sprint 4: technique becomes the autopsy's fifth layer; F1.13 carries objects, materials and skills; heritage-brand and restitution cases in F1.24 and F1.17; the Lineage view opens at F1.1.
- v3.6 — F1 sprints 5 and 6: F1.28a Reconstruction added (its voice and forms to confirm in its spec); each world chapter ends on a living practice and has a named device; Practice Reading; the human-flow rule in F1.13; the state-authored heritage list in F1.3;

35 units. Merged into this register on 2026-09-28 after gate G1. Program unit count about 460.

Research status by module. All modules Open. Sprint order proposed: F1, M2, S2 (the two proofs plus the first studio), then the rest of Foundations and Method, then areas, then venture.

Pass 2

Pass 2 takes one module through the whole workflow with research uncapped, and its first output is a research log, not a lesson.

Workflow for a module (the Module Factory)

1. Research sprint: open questions, sources, world-set candidates, claims sheet, structural findings.
2. Structure update: any adds, splits, merges or re-depths, versioned in §12.
3. Unit specs: for each unit, the essential question, the lab, the lecture element and its conversion, the cases, the Apply for both lanes, the partner voice, the artifact field, the media list.
4. Engine layers: which layers this module switches on, with the formulas and couplings.
5. Copy: unit text in the program voice; argument cards; glossary entries.
6. Prototypes: the module's labs as working pages; engine layers added.
7. Partner prompts: written and tested against the module's units.
8. Checks: accessibility, claims against sources, licences.
9. Your review; revisions; status to Drafted, then Tested.

Proposed first module: F1 Histories of making. It exercises the world set, the Timeline Atlas engine, the free-collection sourcing, the lecture-to-lab conversion for narrative history, and the reference-to-appropriation ethics that the whole program depends on. M2 follows as the second proof.

What I need from you before Pass 2 starts: confirmation of F1 (or another module), and any objections to the depths and unit lists above — comment on the rows themselves.